Jiaju Wu

Peking University

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Research interests Mechanistic Interpretability, Neural Network Architectures, and Learning Dynamics.

Education

Peking University Sep. 2024 – Jun. 2028 (expected)
B.S. in Mathematics, School of Mathematical Sciences **GPA: 3.79 / 4.00**

Selected coursework: Mathematical Analysis I–III; Advanced Algebra I–II; Probability; Mathematical Statistics; Stochastic Processes; Mathematical Foundations of Machine Learning.

Research

Mechanisms of Computation in Looped Transformers Summer 2026 – Present
Shared Weights, Selected Computations: How Looped Transformers Route What Each Loop Does
Jiaju Wu, Yi Hu, Muhan Zhang | Submitted to ICLR 2027
arXiv:2609.39892 | Code and reproduction materials

- Conducted experimental design, implementation, mechanism analysis, and manuscript writing under the supervision of Prof. Muhan Zhang.
- Trained affine state maps to steer frozen looped Transformers on graph-walk tasks; showed that attention-pattern interventions can switch the controlled target, providing causal evidence for attention routing.
- Compared five matched pairs of backbones and found that intermediate supervision changed which transitions the same map family could induce; evaluated composition across successive loops.

Selected Projects

Loss-Curve Prediction Across Learning-Rate Schedules Spring–Summer 2026

- Developed a source-only residual correction for cosine-to-WSD loss prediction, using tangent-space projection to fit a schedule-response coefficient from cosine-schedule losses.
- Improved prediction on all 15 public WSD target curves across 25M, 100M, and 400M models: **30.88% mean relative MAE reduction** over the Multi-Power Law baseline. Audited target-loss leakage and tested projection ablations and transfer across model sizes.

Source Attribution of Mathematical Solutions Spring 2026

- Built lexical and structural features to distinguish human and LLM-generated solutions; compared feature-based classifiers with DistilBERT and evaluated robustness to LLM-generated rewrites.

Grokking on Small Algorithmic Datasets Winter 2025

- Co-developed a PyTorch framework for modular arithmetic and group-operation tasks with Transformer, MLP, LSTM, and GRU baselines; studied delayed generalization across data fractions and optimization settings.

Honors & Service

Gold Medal, Chinese Mathematical Olympiad (CMO) 2023
Lingjun Linghang University Scholarship 2025
Teaching Assistant, Python and AI Fundamentals Summer 2025
Head, Academic Department, SMS Student Union, Peking University 2025–2026

Technical Skills

Python, PyTorch; activation and attention-pattern patching; representation analysis; experimental controls and ablations; reproducible training and evaluation pipelines.